

ATKOOL V3

Software Requirements Specification (SRS)

Version: 3.0 Project Name: ATKOOL Product Type: School ERP, Attendance & Communication Platform

Deployment: SaaS (Multi-School) Platforms:

- Web Application
 - Android Application
 - Progressive Web App (PWA)
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1. Executive Summary

ATKOOL V3 is a complete School ERP platform designed to manage schools, teachers, students, parents, attendance, homework, communication, transportation, reports, examinations, and administrative operations from a single platform.

The primary objective is to provide a connected ecosystem where all modules communicate with each other automatically rather than operating independently.

2. Core Objectives

- Multi-school SaaS platform
 - Parent-teacher communication
 - Student management
 - Staff management
 - Attendance automation
 - Homework management
 - Leave management
 - Transport management
 - Report generation
 - Notification system
 - Role-based access control
 - Mobile-first experience
-

3. User Roles

Super Admin

Responsibilities:

- Manage all schools
 - Create schools
 - Delete schools
 - Suspend schools
 - View analytics
 - Manage subscriptions
 - System-wide reports
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School Admin

Responsibilities:

- Manage school operations
- Manage staff
- Manage students
- Manage fees
- Manage transport
- Manage timetables
- Manage announcements

Restrictions:

- Cannot view SaaS subscription cost details
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Principal

Responsibilities:

- Monitor school performance
 - View attendance
 - Approve requests
 - View reports
 - Staff oversight
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Class Teacher

Responsibilities:

- Manage assigned class
 - Student management
 - Attendance
 - Homework
 - Parent communication
 - Leave approvals
-

Subject Teacher

Responsibilities:

- Manage assigned subjects
 - Attendance
 - Homework
 - Student performance
-

Non-Teaching Staff

Responsibilities:

- Office administration
 - Record management
 - Transport management
 - Administrative tasks
-

Parent

Responsibilities:

- View student details
 - Attendance
 - Homework
 - Reports
 - Notifications
 - Chat with teachers
-

4. Authentication System

Login Methods:

1. Email + Password
2. Mobile Number + Password

Password remains common.

Security:

- JWT Authentication
 - Refresh Tokens
 - Session Management
 - Login History
 - Device Tracking
-

5. OTP Verification System

OTP Delivery Method:

- Gmail SMTP

Used For:

- School deletion
 - Password reset
 - Email verification
 - Account recovery
 - Security verification
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6. School Management Module

Fields:

- School Name
- Logo
- Address
- Phone
- Email
- Website

UDISE Section:

Question: Do you have UDISE ID?

Options:

- Yes
- No

If Yes:

- Enter UDISE ID

Store:

- UDISE Number
 - UDISE Status
-

7. Student Management Module

Student ID System

Replace Roll Number with:

Student ID

Format:

ATK-YYYY-XXXX

Example:

ATK-2026-0001

Student Information

Personal Details:

- Student ID
- Name
- Photo
- Date of Birth
- Gender
- Blood Group

Academic Details:

- Class
- Section
- Admission Number

Guardian Details:

- Father Name
- Mother Name
- Guardian Name
- Primary Contact
- Secondary Contact

Additional Details:

- Address
 - Transport Route
 - Medical Notes
-

8. Class Management Structure

School → Class → Section → Students

Example:

Class 5

- Section A
 - Section B
 - Section C
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Teacher Assignment

Class Teacher: 1 Teacher

Subject Teachers: Multiple Teachers

Important Rule:

Teachers must only see assigned classes.

Example:

Teacher Assigned: Class 5-A

Teacher Dashboard Shows: Class 5-A only

Not: Class 5-B Class 6-A Class 7-C

This fixes the current unstructured class visibility issue.

9. Staff Management Module

Teaching Staff:

- Employee ID
- Name
- Photo
- Qualification
- Experience
- Department
- Assigned Classes
- Assigned Subjects

Non-Teaching Staff:

- Employee ID
 - Designation
 - Department
 - Contact Information
-

10. Attendance Management

Attendance Types:

- Present
- Absent
- Half Day
- Late
- Medical Leave

Features:

- Daily Attendance
- Monthly Attendance
- Yearly Attendance

- Attendance Analytics

Offline Mode:

- Local Storage
 - Auto Sync
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11. Homework Management

Teacher Creates:

Fields:

- Homework Title
- Subject
- Description
- Due Date
- Attachments
- Class
- Section

Workflow:

Teacher ↓ Assign Homework ↓ Parent Notification ↓ Student Submission ↓ Teacher Review

Status:

- Assigned
 - Submitted
 - Reviewed
 - Completed
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12. Leave Management

Parent Applies Leave ↓ Teacher Notification ↓ Approval / Rejection ↓ Attendance Auto Updated ↓ Parent Notification

Leave Types:

- Sick Leave
 - Personal Leave
 - Emergency Leave
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13. ATKOOL CONNECT (Communication Module)

Flagship Feature

Chat Types

Parent ↔ Teacher

Parent ↔ Class Teacher

Teacher ↔ Teacher

Teacher ↔ Principal

Principal ↔ Staff

School Admin ↔ Teachers

School Admin ↔ Parents

Features

- Real-time Chat
 - Read Receipts
 - Typing Indicator
 - Online Status
 - Voice Notes
 - Image Sharing
 - PDF Sharing
 - Search Messages
 - Message History
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Class Based Broadcasts

Example:

Admin selects:

Class 5-A

Message delivered only to:

14. Notification System

Notification Types:

Attendance Homework Leave Examinations Fees Transport Chat Messages Announcements

Real-time Push Notifications.

15. Daily Activities Module

Teacher Logs:

- Today's Lessons
- Classroom Activities
- Homework Summary
- Important Notes

Visible To:

- Parents
 - Principal
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16. Transport Management

Vehicle Details:

- Vehicle Number
- Driver Name
- Driver Contact

Route Details:

- Route Name
- Pickup Points
- Drop Points

Student Assignment:

- Route Mapping

Parent Access:

- Vehicle Details
 - Driver Contact
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17. Fee Management

Class-wise Fee Structure

Fee Types:

- Tuition Fee
- Exam Fee
- Transport Fee
- Miscellaneous Fee

Features:

- Fee Records
- Due Tracking
- Pending Reports
- Fee History

V3: No Online Payment Gateway

Manual Collection Tracking Only

18. Timetable Management

Teacher Timetable

Class Timetable

Subject Timetable

Substitution Management

Conflict Detection

19. Examination Module

Features:

- Exam Creation
- Date Sheet Management
- Marks Entry
- Result Processing

Outputs:

- Report Cards
- Exam Reports

PDF Generation

20. Certificate Management

Generate:

- Bonafide Certificate
- Character Certificate
- Transfer Certificate
- Attendance Certificate

PDF Download

21. Media Center

School Admin Can:

- Upload Photos
- Upload Videos
- Add YouTube Links

Visible To:

- Teachers
 - Parents
 - Students
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22. Reports Module

Attendance Reports

Student Reports

Teacher Reports

Fee Reports

Transport Reports

Leave Reports

Exports:

- PDF
 - Excel
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23. Security Module

Features:

- Email Verification
- Password Reset
- Device Tracking
- Login History
- Audit Logs

Track:

- Attendance Changes
 - Student Updates
 - Staff Updates
 - Leave Approvals
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24. PWA Features

- Install Like Mobile App
- Home Screen Icon
- Offline Access
- Cached Resources

25. Database Collections (MongoDB)

schools

users

roles

students

guardians

teachers

staff

classes

sections

attendance

homeworks

messages

chat_rooms

notifications

fees

transport_routes

vehicles

leave_requests

timetables

exams

results

documents

media

audit_logs

sessions

26. Integration Rules

Attendance → Notifications

Homework → Parent Notifications

Leave → Attendance Updates

Transport → Parent Access

Exams → Report Cards

Chat → Notifications

All modules must remain interconnected.

27. Future Scope (V4)

- AI Timetable Generator
 - Voice Assistant
 - Face Recognition Attendance
 - Online Fee Payment Gateway
 - GPS Tracking
 - Advanced Analytics
 - AI Student Performance Insights
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28. Success Criteria

ATKOOL V3 should function as a complete school operating system where:

- Teachers manage classes efficiently
- Parents remain informed
- Students are monitored properly
- School administration is centralized
- Communication is seamless
- All modules work together automatically
- Data is secure and scalable